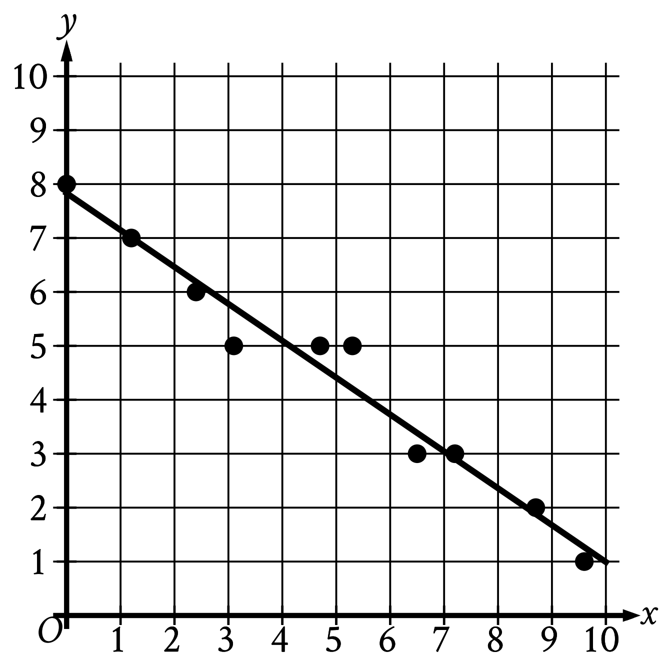
Question ID: 2e74e403

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Medium

Question

In the given scatterplot, a line of best fit for the data is shown.



Which of the following is closest to the slope of this line of best fit?

Answer

A. 7

B. 0.7

C. -0.7

D. -7

Correct Answer: C

Rationale

Choice C is correct. A line of best fit is shown in the scatterplot such that as the value of x increases, the value of y decreases. It follows that the slope of the line of best fit shown is negative. The slope of a line in the *xy*-plane that passes through the points (x_1, y_1) and (x_2, y_2) can be calculated as $\frac{y_2-y_1}{x_2-x_1}$. The line of best fit shown passes approximately through the points $(0, 8)$ and $(10, 1)$. Substituting $(0, 8)$ for (x_1, y_1) and $(10, 1)$ for (x_2, y_2) in $\frac{y_2-y_1}{x_2-x_1}$ yields the slope of the line being approximately $\frac{1-8}{10-0}$, which is equivalent to $\frac{-7}{10}$, or -0.7. Therefore, of the given choices, -0.7 is the closest to the slope of this line of best fit.

Choice A is incorrect. The line of best fit shown has a negative slope, not a positive slope.

Choice B is incorrect. The line of best fit shown has a negative slope, not a positive slope.

Choice D is incorrect and may result from conceptual or calculation errors.